

1. Administrative & Study Identification

Full Study Title: A Single-arm, Multi-center Study of Ruxolitinib for the Treatment of Chinese Patients With Grade II-IV Corticosteroid-refractory Acute Graft Versus Host Disease **Short Title/Acronym:** Study of Efficacy and Safety of Ruxolitinib in Patients With Grade II to IV Steroid-refractory Acute Graft vs. Host Disease **Protocol Number:** CINC424C2416 **NCT Number:** NCT06462469 **Sponsor/Company Name:** Novartis **Investigational Product:** Ruxolitinib **Phase of Development:** Phase 4 **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the efficacy (Overall Response Rate) and safety of ruxolitinib therapy in Chinese adults and adolescents (≥ 12 years old) with Grade II-IV steroid-refractory acute graft versus host disease (SR-aGvHD). * **Secondary Objectives:** To evaluate the durable overall response rate, duration of response, best overall response, non-relapse mortality, overall survival, and long-term safety profile of the investigational product.

2.2 Background & Rationale Grade II-IV acute Graft versus Host Disease (aGvHD) is a severe, life-threatening complication following allogeneic stem cell transplantation (alloSCT). While systemic corticosteroids are the standard first-line treatment, a significant portion of patients become steroid-refractory, leading to high mortality rates. Ruxolitinib, an oral JAK1/2 inhibitor, is being investigated specifically in the Chinese patient population to establish regional efficacy and safety evidence, aiming to address this critical unmet medical need.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target N:** 54 participants * **Number of Sites:** Multi-center (China) * **Estimated Study Start:** 04-Jul-2024 * **Estimated Primary Completion:** 17-Feb-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female Chinese participants aged 12 or older at the time of informed consent. Written informed consent from participant, parent or legal guardian. 2. Have undergone alloSCT from any donor source (matched unrelated donor, sibling, haplo-identical) using bone marrow, peripheral blood stem cells, or cord blood. 3. Clinically diagnosed Grades II to IV acute GvHD as per standard criteria occurring after alloSCT requiring systemic immune suppressive therapy. 4. Able to swallow tablets. 5. Evident myeloid and platelet engraftment (confirmed within 48 hours prior to study treatment start). 6. Confirmed diagnosis of steroid refractory aGvHD defined as participants administered high-dose systemic corticosteroids (methylprednisolone 2 mg/kg/day or equivalent), given alone or combined with calcineurin inhibitors (CNI) and demonstrating progression, failure to achieve at least a partial response, or failure of corticosteroid taper.

4.2 Exclusion Criteria 1. Has received more than one systemic treatment for steroid refractory aGvHD, or participants who received JAK inhibitor therapy for any indication after initiation of current alloSCT conditioning. 2. SR-aGvHD occurring after non-scheduled donor lymphocyte infusion (DLI) administered for pre-emptive treatment of malignancy recurrence. (Note: Scheduled DLIs not for malignancy relapse are permitted). 3. Presence of an active uncontrolled infection including significant bacterial, fungal, viral or parasitic infection requiring treatment. 4. Clinical presentation resembling de novo chronic GvHD or GvHD overlap syndrome with both acute and chronic GvHD features. 5. Failed prior alloSCT within the past 6 months, or presence of relapsed primary malignancy after the alloSCT was performed. 6. Presence of significant respiratory disease, severely impaired renal function, clinically significant or uncontrolled cardiac disease, unresolved cholestatic and liver disorders (not attributable to aGvHD), or disorders/current therapy with medications that interfere with coagulation or platelet function.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Ruxolitinib 10 mg twice daily (BID), given as two 5-mg tablets. * **Route of Administration:** Oral * **Duration of Treatment:** From Day 1 to Week 24 or end of treatment, followed by long-term follow-up until Month 12.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard baseline systemic immune suppressive therapy. * **Prohibited:** Disorders and/or current therapy with medications that interfere with coagulation or platelet function.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Eligibility Assessment, Medical History
Baseline	Day 0	First Dose, Vital Signs, Safety Labs
Interim	Day 28	Primary Efficacy Assessment (ORR evaluation)
End of Treatment	Week 24	Efficacy Assessment, Safety Labs, IP Accountability
End of Study	Month 12	Long-term Safety Follow-up, Survival Status

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Overall Response Rate (ORR) at Day 28 per Investigators. The ORR at Day 28 is defined as the percentage of participants demonstrating a complete response (CR) or partial response (PR) without requirement for additional systemic therapies for an earlier progression, mixed response or nonresponse. * **Measurement Method:** Investigator assessment based on standard clinical GvHD criteria.

7.2 Secondary & Exploratory Endpoints * Durable Overall response rate (ORR) at Day 56 * Duration of Response (DOR) * Best overall response (BOR) * Overall Survival (OS) * Non-Relapse Mortality (NRM) * Safety and tolerability (Incidence of AEs and SAEs)

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard continuous collection via site visits, physical examinations, and laboratory assessments.
- **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the Sponsor safety team.
- **Data Monitoring Committee (DMC):** Internal safety review committee per Novartis standard operating procedures.

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) Set for primary efficacy analysis; Safety Set for all participants who received at least one dose of the study drug.
- **Sample Size Justification:** $N = 54$, adequately powered to evaluate the point estimate of the Day 28 ORR and characterization of the safety profile within the Chinese patient sub-population.
- **Handling of Missing Data:** Participants withdrawing or missing the Day 28 assessment will generally be imputed as non-responders for the primary efficacy analysis.

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Standard clinical monitoring involving on-site/remote visits and source data verification.
 - **Audit Trail:** Data entered into validated Electronic Data Capture (EDC) systems with eCRF locking and comprehensive query resolution.
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11. Study Identifiers & Governance

- **Primary ID:** CINC424C2416
 - **Registry Number:** NCT06462469
 - **Sponsor/Lead Organization:** Novartis
 - **Study Title:** Study of Efficacy and Safety of Ruxolitinib in Patients With Grade II to IV Steroid-refractory Acute Graft vs. Host Disease
 - **Official Regulatory Title:** A Single-arm, Multi-center Study of Ruxolitinib for the Treatment of Chinese Patients With Grade II-IV Corticosteroid-refractory Acute Graft Versus Host Disease
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12. Clinical Framework (SR-aGvHD Specific)

- **Primary Condition/Disease Area:** Steroid-refractory Acute Graft Versus Host Disease (SR-aGvHD)
 - **Diagnosis Threshold:** Clinically diagnosed Grades II to IV acute GvHD occurring after alloSCT requiring systemic immune suppressive therapy
 - **Medical Methodology:** Targeted JAK1/2 Inhibition (Ruxolitinib Therapy)
 - **Optimization Target:** Complete Response (CR) or Partial Response (PR) at Day 28
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13. Study Procedures & Methodology

- **Management Pathway:** Standard clinical pathway for SR-aGvHD incorporating oral ruxolitinib
 - **Education Protocol (HCP):** Protocol specific training / alloSCT site visits
 - **Education Protocol (Patient):** Onsite training for 10 mg BID dosing / self-administration
 - **Quality Audit Mechanism:** Investigator monitoring and clinical response tracking at designated intervals
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14. Observation & Follow-Up Schedule

- **Total Duration:** 12 Months (52 Weeks)
- **Onsite Visit Cadence:** Screening, Baseline, Day 28, Week 24
- **Remote Monitoring Frequency:** Routine safety follow-ups as per protocol following Week 24
- **Checklist Review Interval:** Every 12 weeks during the long-term follow-up period up to Month 12

15. Sign-off & Approval

Role Name	Signature	Date	---	---	---	---
Principal Investigator	_____	[DD-MMM-YYYY]			Clinical	
Operations Lead	_____	[DD-MMM-YYYY]			Lead Statistician	
	_____	[DD-MMM-YYYY]				